

POWERFACTORY

PowerFactory 2024

Technical Reference

WECC Composite Load Model (CMPLDW)

ElmLod,TypLodwecc

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

A *General Load* in conjunction with a *TypLodwecc* type models the WECC dynamic composite load model (CMPLDW) in RMS simulations. Modelling in the other types of calculations is also described in this document.

2 Load-Flow Model

The WECC dynamic composite load model is represented in load flow studies as a static load, defined using a polynomial representation of the voltage dependency and is described by Equation 1. The voltage dependency of loads is only considered if the parameter *Consider Voltage Dependency of Loads* is checked in the *Load Flow Calculation* command dialogue.

$$\begin{aligned} P &= P_0 \cdot \left[aP \cdot \left(\frac{|\underline{u}|}{u_0} \right)^{e_aP} + bP \cdot \left(\frac{|\underline{u}|}{u_0} \right)^{e_bP} + (1 - aP - bP) \cdot \left(\frac{|\underline{u}|}{u_0} \right)^{e_cP} \right] \\ Q &= Q_0 \cdot \left[aQ \cdot \left(\frac{|\underline{u}|}{u_0} \right)^{e_aQ} + bQ \cdot \left(\frac{|\underline{u}|}{u_0} \right)^{e_bQ} + (1 - aQ - bQ) \cdot \left(\frac{|\underline{u}|}{u_0} \right)^{e_cQ} \right] \end{aligned} \tag{1}$$

where aP, bP, cP, aQ, bQ and cQ are the proportional coefficients and e_aP, e_bP, e_cP, e_aQ, e_bQ and e_cQ are the exponents. u_0 is the voltage input parameter (*ElmLod*) and $|\underline{u}|$ is the absolute voltage where the load is connected.

The active and reactive power P_0 and Q_0 are defined as:

$$\begin{aligned} P_0 &= p_{lini} \cdot scale_0 \cdot scale_{zone} \cdot scale_{lf} \\ Q_0 &= q_{lini} \cdot scale_0 \cdot scale_{zone} \cdot scale_{lf} \end{aligned} \tag{2}$$

where p_{lini}, q_{lini} and $scale_0$ are the active power, reactive power and the scaling factor of the load defined in *ElmLod*. Furthermore, $scale_{zone}$ is the zone load scaling factor and $scale_{lf}$ is the load scaling factor defined in the *Load Flow* settings.

By selecting certain values for the proportional and exponential coefficients, different exponential and polynomial models can be defined.

Changing the load type to constant power, current or impedance can be accomplished by modifying the exponential coefficients of the load according to Table 2.1.

Table 2.1: Selection of exponent value for different load model behaviour

Load model type	**Exponential coefficient** (e_aP, e_bP, e_cP)
Constant power	0
Constant current	1
Constant impedance	2

For unbalanced load flow calculations, the WECC composite load model is always represented solely by a positive sequence mode, i.e. the negative- and zero-sequence currents are always set to zero.

2.1 Input Signals

The WECC composite load model can be connected to a QDSL model in order to control its active and reactive power during load flow calculations.

For balanced and unbalanced load flow the power can be controlled through the input signals *pini_set*, *qini_set*, *scale* and *scalephi*. The output power P_0 and Q_0 are then calculated as following:

- When feeder load scaling is enabled:

$$\begin{aligned} P_0 &= pini_set \cdot scale \cdot scale_{lf} \cdot scale_{zone} \\ Q_0 &= qini_set \cdot scalephi \cdot scale_{lf} \cdot scale_{zone} \end{aligned} \tag{3}$$

  where $scale$ and $scalephi$ are determined by the feeder scaling.

- When feeder load scaling is not enabled, first p_0 and q_0 are calculated according to:

$$\begin{aligned} p_0 &= pini_set \cdot scale \cdot scale_{lf} \cdot scale_{zone} \\ q_0 &= qini_set \cdot scale \cdot scale_{lf} \cdot scale_{zone} \end{aligned} \tag{4}$$

  where *scale* is an input signal (initialized to $scale_0$ from the element if not connected). Finally, P_0 and Q_0 are calculated depending if the *scalephi* input is connected or not:

  – *scalephi* input is connected:

$$\begin{aligned} P_0 &= \sqrt{p_0^2 + q_0^2} \cdot cos(scalephi) \\ Q_0 &= \sqrt{p_0^2 + q_0^2} \cdot sin(scalephi) \end{aligned} \tag{5}$$

  – *scalephi* input is not connected:

$$\begin{aligned} P_0 &= p_0 \\ Q_0 &= q_0 \end{aligned} \tag{6}$$

.

3 Short-Circuit Calculation

While the WECC composite load model is an aggregation of static, electronic and induction motor loads, it is still considered as a load model overall. As such, it is not considered in short-circuit calculations according to VDE/IEC, ANSI or IEC 61363.

3.1 Complete Short-Circuit

The WECC composite load model can be modelled either as a constant impedance (Z) or constant current (I) model in the Complete short-circuit method. Z and I are calculated from a preceding load flow analysis.

- Load Model: Impedance, Model 1

  For a load model selected as Impedance, Model 1, the load is modelled as an impedance. Depending on whether the load is capacitive or inductive, either the resistance and inductance or the capacitance and conductance are calculated. The load admittance is calculated as follows:

$$Y_{load} = \frac{I(ldf)}{U(ldf)} \tag{7}$$

  where $I(ldf)$ is the load flow current, $U(ldf)$ is the load flow voltage.

- Load Model: Current Source

  For a load model selected as Current Source, the load current is calculated as follows:

$$Iload = I(ldf) \tag{8}$$

  where $I(ldf)$ is the load flow current, and the load admittance is set to zero $Y_{load} = 0$.

4 Harmonic Analysis

Both the *Impedance, Model 1* load model and the *Current Source* load model are supported.

The *Impedance, Model 1* load model is used in the harmonics calculation essentially as it is used in the short-circuit calculation, as described in Section 3.1.

In harmonics analysis however, the *Impedance, Model 1* load model offers the user the possibility of representing passive loads as *Inductive or capacitive* or *Mixed inductive and capacitive*. In the former case, R and L or C and G are calculated. In the latter case, the proportions of capacitive and inductive reactive power can be specified as a percentage. The active and reactive power is then recalculated into an admittance. Depending on the value of this admittance, the model will be more capacitive or more inductive.

If the *Current Source* load model is used, harmonic current injections can be defined in the element through an Harmonic Sources *TypHmccur*.

5 RMS Simulation

5.1 General Load with *TypLodwecc*, CMPLDW

The WECC composite load model (CMPLDW) is modelled according to [1]. The structure of the load model during RMS simulations is shown in Figure 5.1. The internal buses are not modelled explicitly into the grid, but they are modelled internally in the load model.

The CMPLDW includes a lossless transformer, a shunt capacitor, an equivalent distribution feeder and six loads connected at the internal load bus (feeder receiving end). Additional reactive compensation is added during initialization at the low-side bus and internal load bus in order to compensate for the reactive power drawn by the loads. The six loads are represented by four motor loads, a power electronic load and a static load. *PowerFactory* allows to define each motor as either a three-phase induction motor load or as a three-phase equivalent of single-phase air conditioner performance-based model. Typically, the first three motors represent three-phase motor models and the fourth model represents a single-phase air conditioner model (as per default).

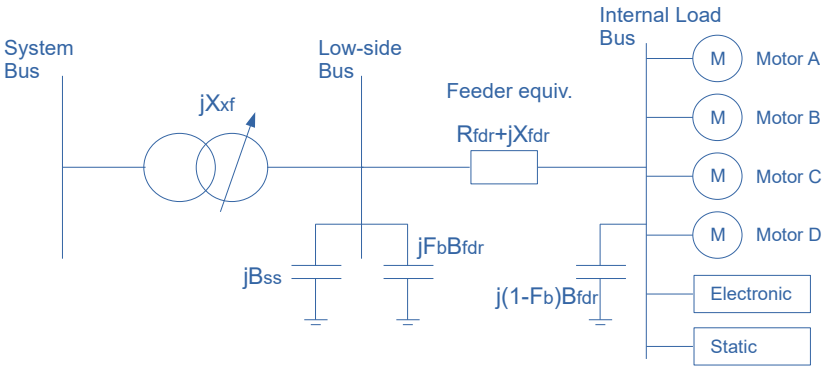


Figure 5.1: WECC composite load model (CMPLDW) structure in RMS simulations

On the element RMS page of the load, under the WECC - CMPLDW tab, it is possible to define:

- parameters related to the on-load tap changer of the transformer
- amount of reactive compensation for the substation shunt capacitor
- the distribution of the additional reactive compensation between low-side bus and internal load bus
- the fraction of the total active power allocated to each load

All remaining parameters are defined in the RMS page of the type *TypLodwecc*. For a detailed explanation of the meaning of each parameter, please refer to [1].

A load with *TypLodwecc* assigned is considered as a simple static load and not as a CMPLDW in RMS simulation if either a) the active power injected during load flow is below the minimum steady-state active power load (parameter $Pmin$); b) the apparent power injected during load flow is below the minimum steady-state apparent power load (% of transformer MVA rating,

parameter $MVAfracmin$). Both these parameters can be defined on the Advanced tab in the RMS page of the type.

In unbalanced RMS simulations, the CMPLDW load is represented as a positive sequence model. Its negative- and zero-sequence currents are always set to zero.

5.1.1 Transformer

The substation transformer is represented according [1]. The lossless transformer has an on-load tap changer (LTC) regulating the low-side voltage and can include a line drop compensation. The LTC is considered differently depending on the value of the *LTC flag* in the element:

- $LTCflag = 0$, the LTC is not active during initialization and simulation
- $LTCflag$ = -1, the LTC is considered only during initialization but remains fixed during simulation
- $LTCflag$ = 1, the LTC is considered both during initialization and simulation

The fixed transformer taps are accounted by modifying the input transformer parameters as:

$$Xxf = Xxf \cdot Tfixhs^2 \tag{9}$$

$$Step = \frac{Step}{Tfixhs} \tag{10}$$

$$Tmin = \frac{Tmin + Tfixls - 1}{Tfixhs} \tag{11}$$

$$Tmax = \frac{Tmax + Tfixls - 1}{Tfixhs} \tag{12}$$

$$Tap = \frac{Tap + Tfixls - 1}{Tfixhs} \tag{13}$$

During simulation, the transformer will tap $Tdel$ seconds after the voltage on the low-side bus is outside the range between $Vmin$ and $Vmax$. Subsequent tap adjustmnents take place at intervals of $Ttap$ seconds.

The $MVAbase$ of the transformer is specified in the type. If the $MVAbase$ is a negative value, it is interpreted as the load power factor, i.e. the actual MVA base is then calculated as:

$$actualMVAbase = \frac{Pload}{|MVAbase|} \tag{14}$$

If the $MVAbase$ is zero, the actual MVA base is then calculated as:

$$actualMVAbase = \frac{Pload}{0.8} \tag{15}$$

where $Pload$ is the active power load injected into the grid by the load from the load flow calculation.

5.1.2 Distribution Feeder

The distribution feeder is represented according [1]. If the voltage at the internal load bus (at feeder receiving end) falls below the value specified by the parameter $Vldmin$, the feeder impedance is decrease in order to keep the internal load bus voltage above $Vldmin$. The parameter $Vldmin$ can be defined in the type, under the Advanced tab in the RMS page.

The additional reactive compensation calculated during initialization in order to compensate for the reactive power drawn by the loads can be fractioned in two parts, at the sending and at the receiving end of the feeder, according to the element parameter Fb.

5.1.3 Static load

The static load is represented according [1]. However, if the voltage at the internal load bus falls below $V1min$ (parameter "Min. voltage for voltage-dependent power reduction", Advanced tab in the RMS page of the type), the active and reactive power components due to exponents $P1e$, $P2e$, $Q1e$, $Q2e$ are adjusted according to Figure 5.2. The adjustment will be performed only if the corresponding exponent is less than 2 in order to avoid convergence issues at low voltages.

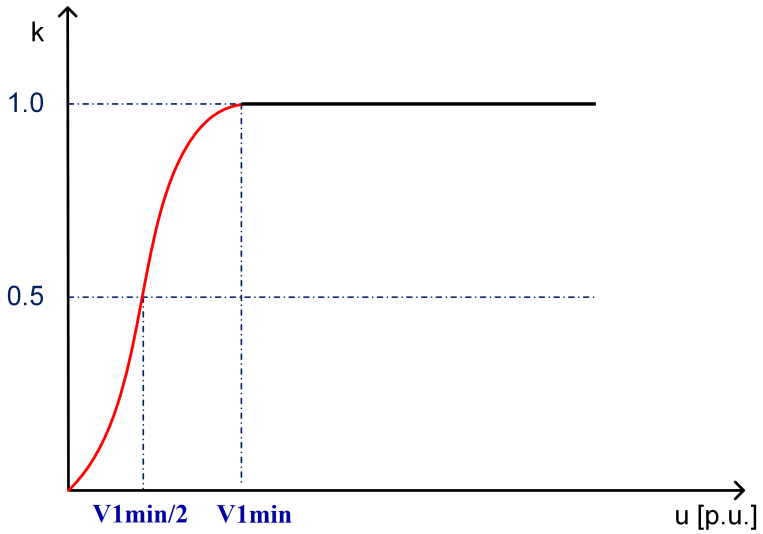


Figure 5.2: Power adjustment for static load

5.1.4 Electronic load

The electronic load is represented according [1].

5.1.5 3-phase motor load

The 3-phase motor load is modelled according to Figure 2 in [1].

5.1.6 1-phase air conditioner motor load

The 1-phase air conditioner motor load is represented according [1]. However, the equations for the two running states have been modified according to:

If $V > Vbrk$ (Run I curve):

$$P = (Po + Kp1 * (V - Vbrk)^{Np1}) \cdot (1 + CmpKpf * \Delta f) \quad (16)$$

$$Q = (Qo + Kq1 * (V - Vbrk)^{Nq1}) \cdot (1 + CmpKqf * \Delta f) \quad (17)$$

If $Vstallbrk < V < Vbrk$ (Run II curve):

$$P = (Po + Kp2 * (Vbrk - V)^{Np2}) \cdot (1 + CmpKpf * \Delta f) \quad (18)$$

$$Q = (Qo + Kq2 * (Vbrk - V)^{Nq2}) \cdot (1 + CmpKqf * \Delta f) \quad (19)$$

If $V < Vstallbrk$ (Stall state curve):

$$P = Gstall \cdot V^2 \quad (20)$$

$$Q = -Bstall \cdot V^2 \quad (21)$$

where $Kp1, Np1, Kq1, Nq1, Kp2, Np2, Kq2, Nq2, Vbrk$ parameters can be defined in the type. $Vstallbrk$ is calculated during initialization in a recursive way as the intersection point between the Run II curve and the stall state curve. Moreover, $Gstall$ and $Bstall$ are calculated as:

$$Gstall + j \cdot Bstall = \frac{1}{Rstall + j \cdot Xstall} \quad (22)$$

If the motor voltage V falls below $Vstall$ for a time longer than $Tstall$ seconds, the motor stays in stalled state. If $Tstall$ is defined as negative, a $Vstall - Tstall$ characteristic (see Figure 5 in [1]) can be defined for the motor on the Advanced tab in the RMS page of the type.

On the Advanced tab in the RMS page of the type, a "Vstall/Vbr adjustment" parameter ($LFAdj$) can be defined. If $LFadj$ is different than zero, the $Vstall$ and $Vbrk$ are ajusted according to:

$$Vstall = Vstall \cdot (1 + LFadj \cdot (LFm - 1)) \quad (23)$$

$$Vbrk = Vbrk \cdot (1 + LFadj \cdot (LFm - 1)) \quad (24)$$

where LFm is the motor loading factor defined in the type.

5.1.7 Initialization

The WECC composite load model is initialized from load flow active and reactive power injection into the grid and bus voltage. During initialization, *PowerFactory* calculates the internal bus voltages, the losses on the feeder and the resulting power flow at the internal load bas where the loads are connected.

The transformer tap is initialized so that the low-side bus voltage is close to $(Vmin + Vmax)/2$.

The amount of additional reactive compensation is calculated iteratively during initialization to compensate for the reactive power drawn from the loads. If the additional reactive compensation is calculated as negative, the amount of fixed shunt capacitive compensation (determined by parameter Bss) is decreased and a warning is printed in the output window.

The active power flow calculated at the internal load bus is then split among the various types of loads proportionally to the element parameters Fma, Fmb, Fmc, Fmd and Fme. If $Fma + Fmb + Fmc + Fmd + Fme$ is less than 1, the remainder of the power is assigned to the static load. If $Fma + Fmb + Fmc + Fmd + Fme$ is greater than 1, they are rescaled so to sum up to 1 and the static load is not considered.

5.2 Load Events

A load event can be defined throughout the course of an RMS simulation via the use of load events. The user must specify a load element, and also a point in time in the simulation for the event to occur. The value of the load can then be altered using the load event. Only a ”Step” load event is supported by the CMPLDW load model.

5.3 Calculation and state variables of the RMS WECC composite load model

Table 5.1: General,substation and distribution feeder

Input Signal	Description	Unit
Tap	Transformer tap	
Rfdr	Feeder R (p.u. of MVA base)	p.u.
Xfdr	Feeder X (p.u. of MVA base)	p.u.
Bfdr	Value of the total feeder reactive compensation	p.u.
Vls	Voltage at substation low-side bus	p.u.
Vld	Voltage at internal load bus	p.u.

Table 5.2: Static load

Input Signal	Description	Unit
Pst	Pst	p.u.
Qst	Qst	p.u.

Table 5.3: Electronic load

Input Signal	Description	Unit
Pel	Pel	p.u.
Qel	Qel	p.u.
Fv1	Fv1	p.u.

Table 5.4: 3-phase motor load

Input Signal	Description	Unit
Pm	Motor P	MW
Qm	Motor Q	Mvar
speed	Speed	p.u.
slip	Slip	
Te	Motor electrical torque	p.u.
Tm	Motor mechanical torque	p.u.
fuv	Fraction of motor not tripped by UV relay	
UVtstart1	tstart1	s
UVtstart2	tstart2	s
fe	Electrical frequency	p.u.
Eppr	Eppr	p.u.
Eppi	Eppi	p.u.
Epr	Epr	p.u.
Epi	Epi	p.u.
Ekr	Ekr	p.u.
Eki	Eki	p.u.
is_r	Stator current, real part (based on Sbase)	p.u.
is_i	Stator current, imaginary part (based on Sbase)	p.u.
ir	Current, real part (based on 1 MVA)	p.u.
ii	Current, imaginary part (based on 1 MVA)	p.u.
Sbase	Sbase	MVA

Table 5.5: 1-phase air conditioner motor load

Input Signal	Description	Unit
Pm	Motor P	MW
Qm	Motor Q	Mvar
Pa	Pa	p.u.
Qa	Qa	p.u.
Pb	Pb	p.u.
Qb	Qb	p.u.
PthA	PthA	p.u.
QthA	QthA	p.u.
PthB	PthB	p.u.
QthB	QthB	p.u.
fthA	fthA	p.u.
fthB	fthB	p.u.
Pth	Pth	p.u.
Qth	Qth	p.u.
ThetaA	ThetaA	p.u.
ThetaB	ThetaB	p.u.
fuv	fuv	
fcn	fcn	
Vmea	Vmea	p.u.
fe	Electrical frequency	p.u.
is_r	Stator current, real part (based on Sbase)	p.u.
is_i	Stator current, imaginary part (based on Sbase)	p.u.
ir	Current, real part (based on 1 MVA)	p.u.
ii	Current, imaginary part (based on 1 MVA)	p.u.
Sbase	Sbase	MVA

6 EMT Simulation

The EMT load model is the same as for the *General Load* with *TypLod* type.

The load model can be selected as *Inductive or capacitive* or as *Mixed inductive and capacitive*. If the latter is selected, the user can also specify the ratio of capacitive/inductive reactive power as a percentage. From this data, the resistance, inductance and capacitance for the load are calculated.

Depending if the option *Variable resistance, inductance and capacitance via input signals* is selected or not, the load can be modelled as a passive load with constant parameters or the resistances, inductances and capacitances per phase can be controlled via external signals.

A References

[1] WECC composite load model specification, April 2021.

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